Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3480**

Title: **INTER-RIVET BUCKLING**

Revision: **1.0**

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1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

Calculation of the inter-rivet buckling load in a fastened joint between a stringer and the skin in a composite stiffened panel loaded in longitudinal compression.

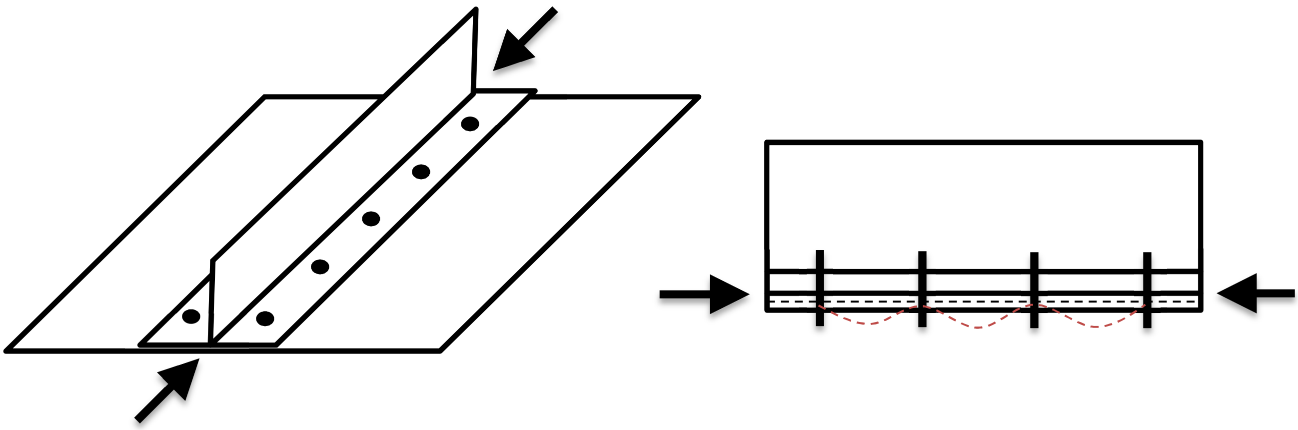

Figure 1. Longitudinal fastened joint with by-pass compression and potential inter-rivet buckling

2 METHOD DESCRIPTION

In a stiffened panel loaded in longitudinal compression, if the stringer is riveted to the skin, the panel may suffer inter-rivet buckling. Even the stringer foot flange may have this failure also, the flange and the fastened joint should be sized to retard inter-rivet buckling beyond the flange crippling strength.

The expression to calculate of the inter-rivet buckling load is a composite panel can be derived from the traditional formula (e.g. Bruhn's chapter C7.14), as follows…

The inter-rivet buckling stress (Bruhn's formula C7.19):

$$\sigma_{ir} = C\cdot\pi^2\cdot E / (L/\rho)^2 = C\cdot\pi^2\cdot E\cdot \rho^2 / L^2 = C\cdot\pi^2\cdot E\cdot (I/A) / L^2 = C\cdot\pi^2\cdot (EI/A) / L^2$$

…where:

- C is the fixity coefficient at rivets locations
- E is the elastic modulus of the material
- L is the length of the column, in this case the rivets pitch (shall be made equal to *p* from now)
- ρ is the radius of gyration of the section, equal to $(I/A)^{0.5}$
- I is the bending moment of inertia of the plate
- A is the cross sectional area of the plate

This expression can be adapted to fibrous composite laminates in a more convenient format by multiplying both terms by the laminate thickness, and replacing *EIt/A* (equal to *EI/b*, i.e. the bending stiffness of the plate per unit width) by its equivalent property in the laminate bending compliance matrix ($EI/b = 1/[D^{-1}]_{11}$):

$$\sigma_{ir}\cdot t = N_{ir} = C\cdot\pi^2 / ([D^{-1}]_{11} \cdot p^2)$$

…where:

- $[D^{-1}]_{11}$ is the first term of the inverted bending stiffness matrix (as B-basis allowable value)

  For laminates unbalanced or unsymmetrical, the "generalized" bending stiffness matrix [D*] shall be used instead of [D] in previous formula:

  $$[D^*] = [D] - [B] \cdot [A]^{-1} \cdot [B]$$

- C is the fixity (or clamping) coefficient at rivets positions. Typical values for C are:

  C = 1.5: countersunk rivets

  C = 3.0: universal rivets

  C = 4.0: flat head rivets

- p is the pitch, or distance, between rivets

The strength check shall be done comparing this allowable strength vs the applied compression, in terms of axial load-flow (or even in terms of apparent stress). Making this comparison in terms of strains is wrong as, due to the Poisson effect, it is possible to have compression longitudinal strain in the joint with the panel subjected exclusively to transverse tension load, which would not produce any inter-rivet buckling.

(*) Kassapoglou's book proposes a similar formula for the inter-rivet buckling strength (see Ref. 3 chapter §8.7). It shows also the condition for selecting the maximum rivets pitch, in fastened flanges, in order to prevent inter-rivet buckling below the crippling of the flange. However, its formulation includes the $[D]_{11}$ term of the laminate flexure stiffness matrix, instead of the $1/[D^{-1}]_{11}$ value proposed in this document (as equivalent to the laminate-beam stiffness). Kassapoglou derives his expression from the laminated plates theory instead of from the Euler's beam theory, as herein. This discrepancy shall be investigated.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
A	mm^2	Area of the cross section
[A]	N/mm	In-plane stiffness matrix of a composite laminate
B	mm	Plate width
[B]	N	Matrix defining the laminate shear-bending coupling
C	-	Fixity coefficient
[D]	Nmm	Bending stiffness matrix of a composite laminate
E	MPa	Young modulus
I	mm^4	Bending moment of inertia of the cross section
N_{ir}	N/mm	Inter-rivet buckling strength as load-flow (force per unit width)
p	mm	Rivets pitch
t	mm	Thickness
ρ	mm	Radius of gyration of the section, equal to $(I/A)^{0.5}$

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[Robert M. Jones]	2nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 3	[C. Kassapoglou]	1st Ed.	Design and Analysis of Composite Structures

SOFTWARE

Program	***Version***	***Description***
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures

Table 1. *Software programs referenced in this chapter*

DISTRIBUTION LIST

Department	***Name***	***Full Doc.***	***Cover Page***
Stress Methods & Optimisation (TEASP-TL3)	Fernass Daoud		X
Fatigue & Damage Tolerance GET (TEASA-TL1)	Javier Gomez-Escalonilla Martin		
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Fatigue & Damage Tolerance MAN (TEASA-TL7)	Patrick Schneider		

Table 2. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.2 – 1.0 01/11/2018	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Analysis (TEASA)	Miguel Ángel Calero 26/06/2019	Bernhard Mayer 26/06/2019
Stress Methods (TEASP-TL3)	G. Baños 01/11/2018	Florian Glock 26/06/2019

Table 4. *Approval signatures table for last revision of this chapter*